

Term 3 Learning Evidence: C6 and C7 Solutions

Contents

- 1 (6) Fuel-Inflation Claim Decision (C6)
- 2 (6) Unemployment-Duration Reform Decision (C6)
- 3 (7) Minimum Sample Size for Export Price Evidence (C7)
- 4 (7) Minimum Sample Size for Delivery-Time Evidence (C7)
- 5 (7) Minimum Sample Size for Spending-Deviation Evidence (C7)

Evaluated criteria

- C6. Interprets decisions from hypothesis tests in terms of context, including p-value comparisons and error types.
- C7. Uses one-sample mean z -tests with known σ to find minimum sample sizes required to reject H_0 .

1 (6) Fuel-Inflation Claim Decision (C6)

A city transport authority tests whether mean monthly fuel inflation is above the 4.0% planning benchmark:

$$H_0 : \mu = 4.0, \quad H_1 : \mu > 4.0, \quad \alpha = 0.05.$$

The report provides $z_{obs} = 2.05$ and $p\text{-value} = 0.020$.

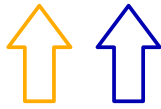
Questions/Tasks.

- a) (2) State the decision rule and determine whether to reject or fail to reject H_0 .
 - b) (2) Interpret the statistical conclusion in context.
 - c) (2) Identify the relevant error type and explain its practical implication.
-
- a) (2) State the decision rule and determine whether to reject or fail to reject H_0 .

Decision rule:

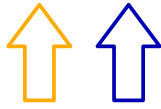
$$\text{Reject } H_0 \text{ if } p\text{-value} < \alpha, \quad \text{Fail to reject } H_0 \text{ if } p\text{-value} \geq \alpha.$$

Here, $p\text{-value} = 0.020 < 0.05 = \alpha$, so we **reject** H_0 .



b) (2) Interpret the statistical conclusion in context.

Because we reject H_0 , the data provide sufficient evidence (at the 5% significance level) that mean monthly fuel inflation is **greater than 4.0%**.



c) (2) Identify the relevant error type and explain its practical implication.

Since the decision is to reject H_0 , the relevant risk is a **Type I error**. In context, this would mean concluding that average fuel inflation exceeds 4.0% when, in reality, the true mean is still 4.0%.



2 (6) Unemployment-Duration Reform Decision (C6)

A labor ministry evaluates whether a reform reduced average unemployment duration below 11 weeks:

$$H_0 : \mu = 11, \quad H_1 : \mu < 11, \quad \alpha = 0.10.$$

The report provides $z_{obs} = -0.85$ and $p\text{-value} = 0.198$.

Questions/Tasks.

- a) (2) State the decision rule and determine whether to reject or fail to reject H_0 .
- b) (2) Interpret the statistical conclusion in context.
- c) (2) Identify the relevant error type and explain its practical implication.

a) (2) State the decision rule and determine whether to reject or fail to reject H_0 .

Decision rule:

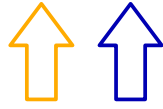
$$\text{Reject } H_0 \text{ if } p\text{-value} < \alpha, \quad \text{Fail to reject } H_0 \text{ if } p\text{-value} \geq \alpha.$$

Here, $p\text{-value} = 0.198 \geq 0.10 = \alpha$, so we **fail to reject** H_0 .



b) (2) Interpret the statistical conclusion in context.

At the 10% significance level, there is insufficient evidence to conclude that the reform reduced mean unemployment duration below 11 weeks.



c) (2) Identify the relevant error type and explain its practical implication.

Since the decision is to fail to reject H_0 , the relevant risk is a **Type II error**. In context, this means a real reduction below 11 weeks could exist, but the ministry might not detect it.



3 (7) Minimum Sample Size for Export Price Evidence (C7)

For export prices, test:

$$H_0 : \mu = 120, \quad H_1 : \mu > 120,$$

with known $\sigma = 20$, observed $\bar{x} = 124$, and $\alpha = 0.05$.

Questions/Tasks.

a) (3) Derive $z(n)$, the rejection inequality, and find $n_{\min}$.

b) (2) Interpret $n_{\min}$ in context.

a) (3) Derive $z(n)$, the rejection inequality, and find $n_{\min}$.

Let μ be the population mean export price.

$$z(n) = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{124 - 120}{20/\sqrt{n}} = 0.2\sqrt{n}.$$

For a right-tailed test at $\alpha = 0.05$, the critical value is $z_\alpha = 1.645$, and rejection requires:

$$0.2\sqrt{n} \geq 1.645 \Rightarrow \sqrt{n} \geq \frac{1.645}{0.2} = 8.225 \Rightarrow n \geq 67.65.$$

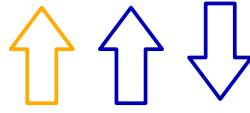
Thus,

$$n_{\min} = \left\lceil \left(\frac{1.645}{0.2} \right)^2 \right\rceil = 68.$$



b) (2) Interpret $n_{\min}$ in context.

At least 68 export-price observations are needed so that this sample mean difference is large enough (relative to σ) to reject H_0 at the 5% level.



4 (7) Minimum Sample Size for Delivery-Time Evidence (C7)

For delivery-time records, test:

$$H_0 : \mu = 18, \quad H_1 : \mu < 18,$$

with known $\sigma = 4$, observed $\bar{x} = 17$, and $\alpha = 0.10$.

Questions/Tasks.

a) (3) Derive $z(n)$, the rejection inequality, and find $n_{\min}$.

b) (2) Interpret $n_{\min}$ in context.

a) (3) Derive $z(n)$, the rejection inequality, and find $n_{\min}$.

Let μ be the population mean delivery time (days).

$$z(n) = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{17 - 18}{4/\sqrt{n}} = -0.25\sqrt{n}.$$

For a left-tailed test at $\alpha = 0.10$, rejection requires $z(n) \leq -z_\alpha = -1.282$:

$$-0.25\sqrt{n} \leq -1.282 \Rightarrow \sqrt{n} \geq \frac{1.282}{0.25} = 5.128 \Rightarrow n \geq 26.30.$$

Thus,

$$n_{\min} = \left\lceil \left(\frac{1.282}{0.25} \right)^2 \right\rceil = 27.$$



b) (2) Interpret $n_{\min}$ in context.

At least 27 observations are required to obtain enough evidence to reject H_0 and support that mean delivery time is below 18 days at the 10% level.



5 (7) Minimum Sample Size for Spending-Deviation Evidence (C7)

For weekly household imported-food spending, test:

$$H_0 : \mu = 75, \quad H_1 : \mu \neq 75,$$

with known $\sigma = 12$, observed $\bar{x} = 78$, and $\alpha = 0.05$.

Questions/Tasks.

a) (3) Derive $|z(n)|$, the two-tailed rejection inequality, and find $n_{\min}$.

b) (2) Interpret $n_{\min}$ in context.

a) (3) Derive $|z(n)|$, the two-tailed rejection inequality, and find $n_{\min}$.

Let μ be the population mean weekly imported-food spending.

$$|z(n)| = \left| \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \right| = \left| \frac{78 - 75}{12/\sqrt{n}} \right| = 0.25\sqrt{n}.$$

For a two-tailed test with $\alpha = 0.05$, the critical value is $z_{\alpha/2} = z_{0.025} = 1.96$, and rejection requires:

$$0.25\sqrt{n} \geq 1.96 \Rightarrow \sqrt{n} \geq \frac{1.96}{0.25} = 7.84 \Rightarrow n \geq 61.47.$$

Thus,

$$n_{\min} = \left\lceil \left(\frac{1.96}{0.25} \right)^2 \right\rceil = 62.$$



b) (2) Interpret $n_{\min}$ in context.

A minimum of 62 surveyed households is needed to reject H_0 at the 5% level and conclude that mean spending differs from 75 (in either direction).

